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Corso di Laurea in Informatica

Appunti Causal Networks

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Chapter 1

Introduction to Causal Networks

With the term Artificial Intelligence (**AI**) we refer to the intelligence demonstrated by machines that mimic cognitive functions that humans associate with other human minds, such as learning and problem-solving. Machine learning (**ML**) is a subset of AI compound by algorithms and statistical models that computer systems use to perform specific tasks without using explicit instructions, relying on patterns and inference instead. There are 3 types of ML: supervised learning, self-supervised (or unsupervised learning) and reinforcement learning. Deep Learning (**DL**) is a subset of ML that uses artificial neural networks to model and understand complex patterns and relationships in data. Anyway, everything is based on curve fitting, linear or non-linear, and the goal is to minimize the error between the predicted output and the actual output. For this reason, studying the causation is important to make sense of data, to understand how to have an effect on a system and to guide actions and policies. In the Figure 1.1 you can see the relationship between AI, ML and DL.

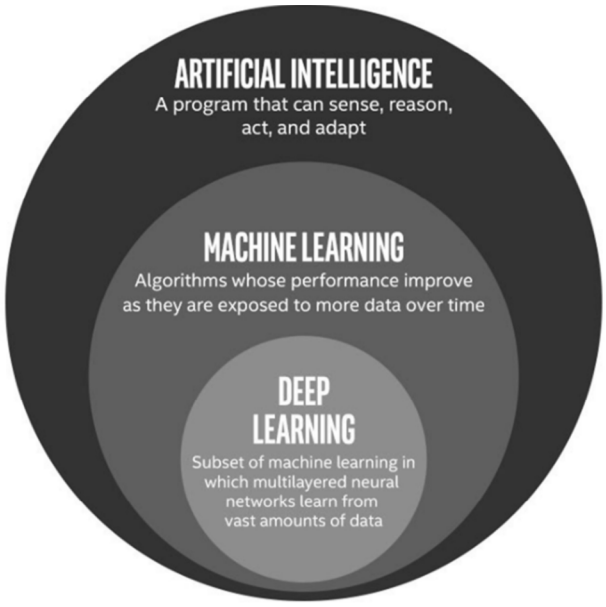


Figure 1.1: The set of AI, ML and DL.

1.1 Simpson's Paradox

Statistical phenomenon in which an observed trend in different groups disappears or reverses when the groups are combined, so this can lead to an evaluation error. So data alone aren't enough to fully understand a system and to make decisions (**correlation is not causation**). In Fig. 1.2 there is an example, if we consider only the combined data the drug appears to be harmful: only 78% of the patients recovered compared to 83% of the patients that didn't take the drug, but if we consider the data divided by gender, we can see that the drug is actually beneficial for both men and women.

Table 1.1 Results of a study into a new drug, with gender being taken into account

	Drug			No Drug		
	patients	recovered	% recovered	patients	recovered	% recovered
Men	87	81	93%	270	234	87%
Women	263	192	73%	80	55	69%
Combined data	350	273	78%	350	289	83%

Figure 1.2: Simpson's Paradox - Example.

1.1.1 The Pitfall of Blind Data Segregation

A common mistake is assuming that segregating data by any available variable always yields the correct decision.
Indeed, as shown in the following example (Fig 1.3) is not correct to just segregate the data whenever is possible. In this example we were considering the effect of a drug on the recovery of patients with an High BP by lowering the BP, unfortunately the drug has also a toxic effect. The segregated data shows that without taking the drug the recovery rate is higher for both groups (Low BP and High BP). However considering only the subpopulations in this case would introduce a bias as we would only see the drug's toxic effect. In general population, the drug might improve recovery rates because it has an effect on BP, a person with High BP has 73% of recovery rate without taking the drug, but if he takes the drug he could lower his BP and recover with 87% of probability. If we consider only the subpopulations

Table 1.2 Results of a study into a new drug, with posttreatment blood pressure taken into account

	No Drug			Drug		
	patients	recovered	% recovered	patients	recovered	% recovered
Low BP	87	81	93%	270	234	87%
High BP	263	192	73%	80	55	69%
Combined data	350	273	78%	350	289	83%

Figure 1.3: Simpson's Paradox - Second Example.

1.2 Ladder of Causation

Level of Association

Prediction based on passive observations, for instace an marketing director asking if a customer who bought a toothpase is likely to buy a toothbrush.

Level of Intervention

Actionable predictions based on interventions, for instance, a marketing director asking if a customer is likely to buy a toothbrush if offered a discount.

Level of Counterfactual

Predictions based on hypothetical outcomes of events that happened in the past. For instance, a marketing director asking if a customer would have bought a toothbrush if he had not bought a toothpaste.

Chapter 2

Potential Outcomes Framework

2.1 Individual Treatment Effect

SISTEMARE FORMULE, HO UNA MAREA DI FORMULE RIPETUTE LE ASSUMPTION POSSONO DIVENTARE DELL DEF Before going into the formal definition of the individual treatment effect, we need to introduce some notation:

- X it's the random variable for **Treatment** (most of the time binary)
- Y it's the random variable for **Outcome**, also called the **observed outcome** (most of the time binary)
- Z it's a set of random variables for **Covariates**
- $Y(x)$ it's the **potential outcome** if the treatment $X = x$ is applied
- X_i, Y_i, Z_i are the realizations of the random variables X, Y, Z for the i-th individual in the population

In general, the uppercase letters are used to denote random variables, while the lowercase letters are used to denote their realizations.
For example, consider the case in which we have:

- Happiness as the outcome variable Y
- Dog as the treatment variable X
- Y(1) the potential outcome of happiness if the individual gets a dog (X=1)
- Y(0) the potential outcome of happiness if the individual does not get a dog (X=0)

we could have the scenarios observed in 2.1

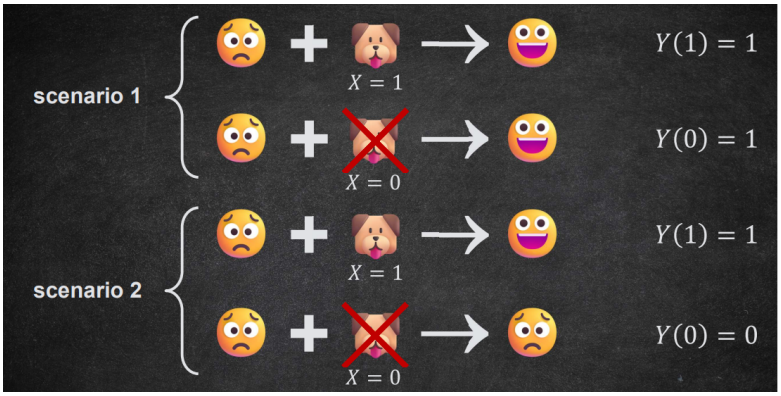


Figure 2.1: Individual Treatment Effect

It's important to point out that:

- $Y(x)$ is distinct from Y because not all potential outcomes are observed, although all potential outcomes could be observed.
- The actually observed potential outcome depends on the given value x of treatment X.

Of course, in reality the population is composed of many individuals or units, every one of which is associated with one or more variables (Z).

Definition 1 (Individual treatment effect (ITE)) *for the ith individual is defined as the difference between the potential outcome for the treatment $X = 1$ and the potential outcome for the treatment $X = 0$:*

$$\tau_i \triangleq Y_i(1) - Y_i(0) \tag{2.1}$$

It is one of the main quantities of interest in causal inference. Y(x) is a random variable, while $Y_i(x)$ is treated as non-random because it is restricted to a specific individual whose potential outcomes are deterministic. This is a theoretical/conceptual definition because, as we said before, we can only observe one of the two potential outcomes for each individual. This leads us to the **Fundamental Problem of Causal Inference**.

2.2 The Fundamental Problem of Causal Inference

If we consider the example of the dog and happiness, we can see that an individual can't get a dog, observe Y(1), give the dog away and then observe Y(0). The second observation will be influenced by all the action taken between the two observations. [1] There are others two definitions:

- **Counterfactual**: potential outcomes that aren't (and can't be) observed
- **Factual**: potential outcomes that are observed

Note that there are no counterfactuals or factuals until the outcome is observed.
What helps us solve this problem is the average of treatment effect.

[1] ML cares only about predicting Y, so ML does not have to deal with the fundamental problem of causal inference.

Definition 2 (Average Treatment Effect (ATE)) *obtained by taking an average over the ITEs*

$$\tau \triangleq E[\tau_i] = E[Y_i(1) - Y_i(0)] = E[Y(1) - Y(0)] \tag{2.2}$$

Let now us consider a population of 6 individuals, with the data shown in Table 2.1. The fundamental problem of causal inference can be seen as a **Missing Data Problem** where the question marks (?) mean that we do not observe the value for the corresponding cell. So

i	X	Y	$Y(1)$	$Y(0)$	$Y(1) - Y(0)$
1	0	0	?	0	?
2	1	1	1	?	?
3	1	0	0	?	?
4	0	0	?	0	?
5	0	1	?	1	?
6	1	1	1	?	?

Table 2.1: Potential Outcomes and Missing Data

the problem here is that we cannot compute the ATE (or ACE) directly because of missing counterfactual values. Furthermore, we cannot simply rely on the **Associational Difference**:

$$\mathbb{E}[Y|X = 1] - \mathbb{E}[Y|X = 0]$$

because it compares two distinct subpopulations ($X = 1$ vs $X = 0$) rather than the same individuals under both conditions, indeed it isn't a causal quantity. However we can rewrite the formula 2.2 by using linear properties of expectation[2]:

$$\tau \triangleq \mathbb{E}[\tau_i] = \mathbb{E}[Y(1) - Y(0)] = \mathbb{E}[Y(1)] - \mathbb{E}[Y(0)] \tag{2.3}$$

The *associational difference* and the formulation of the ATE we just obtained are not equal due to **confounding**. In Figure 2.2 is depicted how the covariate Z confounds the effect of X on Y, if we pay attention is also the graph of variables of Simpson's paradox example showed in Cap 1.1.

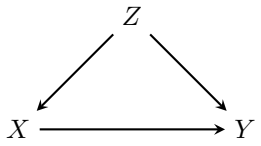


Figure 2.2: Causation between variables

In conclusion we can't **in general** compare the *associational difference* with the *ATE* due to confounding. However there are some *assumption* under which the equality holds true:

$$\tau \triangleq \mathbb{E}[\tau_i] = \mathbb{E}[Y(1)] - \mathbb{E}[Y(0)] = \mathbb{E}[Y|X = 1] - \mathbb{E}[Y|X = 0] \tag{2.4}$$

[2]For any random variables A and B and constants $a, b \in \mathbb{R}$: $\mathbb{E}[aA + bB] = a\mathbb{E}[A] + b\mathbb{E}[B]$ regardless of whether A and B are independent or dependent

2.3 Assumptions for ATE

As discussed, the Fundamental Problem of Causal Inference presents itself as a **missing data problem**, preventing us from directly computing the ATE. To bridge the gap between theoretical potential outcomes and observable data, we must rely on a set of core identifying assumptions centered around the concept of ignorability.

Definition 3 (Ignorability / Exchangeability) *Ignorability states that the potential outcomes are independent of the treatment assignment:*

$$(Y(1), Y(0)) \perp\!\!\!\perp X$$

It means that we can ignore how individuals ended up selecting treatment X. In terms of populations, the groups are exchangeable *in the sense that if their treatment assignments were swapped, the new treatment group would observe the same outcome distribution as the old treatment group, and vice versa.*

So we don't consider the causality between X and Z as shown in Fig 2.3

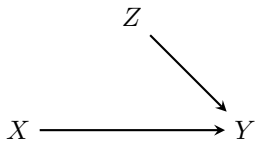


Figure 2.3: Causation between variables

This assumption alone allows us to use the formula 2.4[3] and it also guarantees that the groups are comparable ($X = 1$ and $X = 0$ are the same in all relevant aspects other than the treatment X).
Another aspect in which this assumption is crucial is that it allows us to **identify causal effects**, in other words, it satisfies the property of **Identifiability**:

Definition 4 (Identifiability) *A causal quantity (e.g., $\mathbb{E}[Y(x)]$) is identifiable if it can be computed exclusively from purely statistical quantities derived from observational data (e.g., $\mathbb{E}[Y \mid X = x]$).*

However, making this assumption in observational studies is often **unrealistic** (or a strong identifying assumption). In practice, unobserved confounders frequently invalidate full ignorability, which is why conditioning on observed covariates Z (conditional ignorability) becomes necessary.

Definition 5 (Unconfoundedness / Conditional Exchangeability) *conditional on a set of observed covariates, the treatment is independent of the potential outcomes:*

$$(Y(1), Y(0)) \perp\!\!\!\perp X|Z \tag{2.5}$$

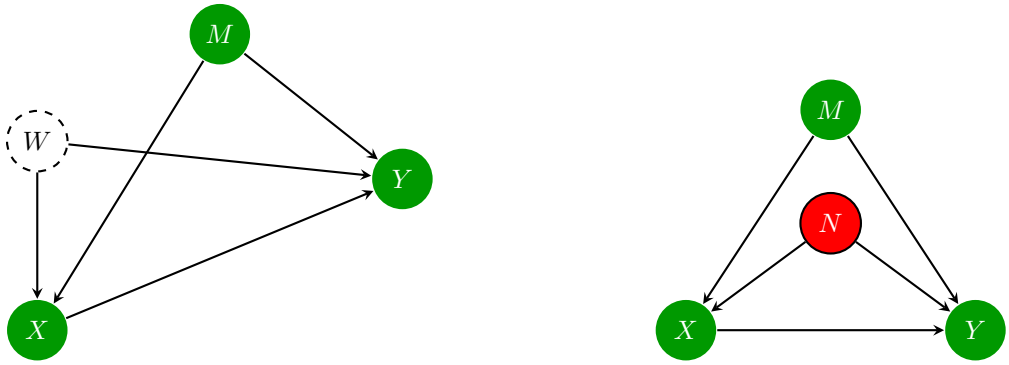
This means that although the treatment and potential outcomes may be unconditionally associated (due to confounding), within levels of Z, they are not associated. The *non-causal association*

[3]Strictly speaking, the weaker assumption of *Mean Ignorability* (or mean exchangeability, constraining only the first moment of the distribution: $\mathbb{E}[Y(1) \mid X = x] = \mathbb{E}[Y(1)]$) would be sufficient for the ATE. However, it is standard practice to assume complete (full) ignorability as defined here.

$X \leftarrow Z \rightarrow Y$ is blocked at Z by conditioning on Z (X and Y are d-separated by Z).
With this new assumption the 2.4 becomes:

$$\tau \triangleq \mathbb{E}[\tau_i] = \mathbb{E}[Y(1)|Z] - \mathbb{E}[Y(0)|Z] = \mathbb{E}[Y|X=1,Z] - \mathbb{E}[Y|X=0,Z] \tag{2.6}$$

There may be some unobserved confounders (W as shown in Figure 2.4a) that are not part of the observed covariate set $Z = \{M\}$, meaning that **unconfoundedness is violated**. Fortunately, this is typically not an issue in ideal **randomized experiments**. However, it remains a critical challenge that we must always account for when working with **observational data**. Intuitively, the best practice is to observe and adjust for as many relevant covariates in Z as possible to support the unconfoundedness assumption. However, as we will explore in subsequent lectures, conditioning on more covariates does not automatically yield less biased causal estimates. In fact, adjusting for the "wrong covariates" (e.g., conditioning on N in $Z = \{M, N\}$ as illustrated in Figure 2.4b) can actually introduce or increase estimation bias. Specifically, adjusting for M provides unbiased estimates, whereas adjusting for N results in biased estimates. Conditioning



(a) Unobserved Confounder W (b) Wrong Covariate N

Figure 2.4: Comparison between an unobserved confounder W and a problematic covariate adjustment N.

on many covariates is attractive for achieving unconfoundedness, but it can be detrimental for another reason that has to do with positivity.

Definition 6 (Positivity / Overlap / Common Support) *For all values $\boldsymbol{z}$ of covariates $\boldsymbol{Z}$ present in the population of interest (i.e., $\boldsymbol{z}$ such that $P(\boldsymbol{Z} = \boldsymbol{z}) > 0$):*

$$0 < P(X = 1 \mid \boldsymbol{Z} = \boldsymbol{z}) < 1 \tag{2.7}$$

In other words, is the condition that all subgroups of the data with different value for covariates have some probability of receiving any value of treatment.

Remark 1 (Positivity Violation Leads to Division by Zero) *if we rewrite the right-hand side of*

$$\mathbb{E}[Y(1) - Y(0)] = \mathbb{E}_{\mathbf{Z}}[\mathbb{E}[Y \mid X = 1, \mathbf{Z}] - \mathbb{E}[Y \mid X = 0, \mathbf{Z}]]$$

in the case of discrete covariate variables $\mathbf{Z}$*, to obtain:*

$$= \sum_{\mathbf{z}} P(\mathbf{Z} = \mathbf{z}) \left(\sum_{y} y\, P(Y = y \mid X = 1, \mathbf{Z} = \mathbf{z}) - \sum_{y} y\, P(Y = y \mid X = 0, \mathbf{Z} = \mathbf{z}) \right)$$

$$= \sum_{\mathbf{z}} P(\mathbf{Z} = \mathbf{z}) \left(\sum_{y} y \frac{P(Y = y, X = 1, \mathbf{Z} = \mathbf{z})}{P(X = 1 \mid \mathbf{Z} = \mathbf{z}) P(\mathbf{Z} = \mathbf{z})} - \sum_{y} y \frac{P(Y = y, X = 0, \mathbf{Z} = \mathbf{z})}{P(X = 0 \mid \mathbf{Z} = \mathbf{z}) P(\mathbf{Z} = \mathbf{z})} \right)$$

As we can observe from the denominators in the final expression, if the positivity assumption is violated for some value $\mathbf{z}$ *(meaning* $P(X = 1 \mid \mathbf{Z} = \mathbf{z}) = 0$ *or* $P(X = 0 \mid \mathbf{Z} = \mathbf{z}) = 0$*), the expression results in a division by zero, making the causal effect undefined for that stratum. This makes positivity essential to define causal effect.*

The last two assumptions are part of the **Stable Unit Treatment Value Assumption (SUTVA)**, that is satified if unit i's outcome Y_i is a function of unit i's treatment X_i. SUTVA is a combination of two assumptions:

Definition 7 (No Interference) *The outcome* Y_i *of each unit* i *is unaffected by anyone else's treatment* X_j *for* $j \neq i$*.*

$$Y_i(x_1, x_2, ..., x_i, ..., x_n) = Y_i(X_i) \tag{2.8}$$

Definition 8 (Consistency) *If the treatment is* X*, then the observed outcome* Y *is the potential outcome under treatment. Formally*

$$X = x \implies Y = Y(x) \text{ or equivalently } Y = Y(X) \tag{2.9}$$

The consistency assumption might seem trivial, but it can be violated in practice. For example, if X = "get a dog","don't get a dog", the potential outcome Y("get a dog") cuold be 1 because i needed an energetic dog, but if i got a lazy dog the potential outcome Y("get a dog") could be 0. In this case, the treatment is not well-defined and the consistency assumption is violated.
With all this assumptions we can derived the **Adjustment Formula** starting from the ATE 2.2

Definition 9 (Adjustment Formula) *Given the assumptions of unconfoundedness, positivity, consistency, and no interference, we can identify the ATE:*

$$\tau \triangleq E[\tau_i] = \mathbb{E}_{\mathbb{Z}}[\mathbb{E}[Y|X = 1, Z] - \mathbb{E}[Y|X = 0, Z]] \tag{2.10}$$

Dimostration 1

$$\begin{aligned}
\mathbb{E}[Y(1) - Y(0)] &= \mathbb{E}_{\mathbb{Z}}[\mathbb{E}[Y(1) - Y(0)|Z]] \ (\textit{linearity of expectation}) \\
&= \mathbb{E}_{\mathbb{Z}}[\mathbb{E}[Y(1)|Z] - \mathbb{E}[Y(0)|Z]] \ (\textit{law of iterated expectations}) \\
&= \mathbb{E}_{\mathbb{Z}}[\mathbb{E}[Y(1)|X = 1, Z] - \mathbb{E}[Y(0)|X = 0, Z]] \ (\textit{unconfoundedness and positivity}) \\
&= \mathbb{E}_{\mathbb{Z}}[\mathbb{E}[Y|X = 1, Z] - \mathbb{E}[Y|X = 0, Z]] \ (\textit{consistecy})
\end{aligned}$$

2.4 Core Terminology: Estimands, Estimators, and Estimates

Definition 10 (Estimand) *An estimand is a quantity that we want to estimate.*

Definition 11 (Estimate) *An estimate is an approximantion of some estimand, which we get using the data*

Definition 12 (Estimator) *An estimator is a function that maps a dataset to an estimate of the estimand.*

Most of the time, estimators use a model to estimate quantities (**Model-Assisted Estimators**)

Definition 13 (Estimation) *An estimation is the process of using an estimator to compute an estimate of an estimand.*

Definition 14 (Causal Estimand) *A causal estimand is any estimand that contains a potential outcome in it.*

Definition 15 (Statistical Estimand) *A statistical estimand is any estimand that does not contain a potential outcome in it.*

For example, in

$$\mathbb{E}[Y(1) - Y(0)] = \mathbb{E}_{\mathbf{Z}}[\mathbb{E}[Y \mid X = 1, \mathbf{Z}] - \mathbb{E}[Y \mid X = 0, \mathbf{Z}]]$$

$\mathbb{E}[Y(1) - Y(0)]$ is the **Causal Estimand** and $\mathbb{E}_{\mathbf{Z}}[\mathbb{E}[Y \mid X = 1, \mathbf{Z}] - \mathbb{E}[Y \mid X = 0, \mathbf{Z}]]$ is the **Statistical Estimand**. So using the defs 4, 14, and 15 we can define:

Definition 16 (Identification) *The process of moving from a causal estimand to an equivalent statistical estimand.*

In Figure 2.5 is depicted the process from a causal estimand to an estimate.



Figure 2.5: Relationship between causal estimands, statistical estimands, and final estimates.

2.5 A Practical Application: Estimating the ATE of Sodium Intake

Let us examine an epidemiological study analyzing the causal effect of daily sodium intake (X) on systolic blood pressure (Y). Since sodium intake is a continuous variable, we binarize it for simplicity:

- $X = 1$ if daily sodium intake ≥ 3.5 gr.
- $X = 0$ if daily sodium intake < 3.5 gr.

The dataset also includes individual-level covariates $\mathbf{Z} = \{\text{Age, Amount of protein in the urine}\}$. Because this example relies on a simulation study, the true underlying average treatment effect is known: $\mathbb{E}[Y(1) - Y(0)] = 1.05$.

2.5.1 Model-Assisted Estimation via Linear Regression

To compute the estimate, we proceed operationally through the following steps:

1. We assume consistency, positivity, and unconfoundedness given $\mathbf{Z}$.
2. We take the outer expectation over $\mathbf{Z}$ in the adjustment formula and replace it with an empirical mean over the n samples in the dataset:

$$\frac{1}{n}\sum_{i=1}^{n}[\mathbb{E}[Y \mid X=1, \mathbf{Z}=\mathbf{z}_i] - \mathbb{E}[Y \mid X=0, \mathbf{Z}=\mathbf{z}_i]]$$

3. We fit a machine learning model to estimate the conditional expectation $\mathbb{E}[Y \mid x, \mathbf{z}]$. Specifically, we use **linear regression** (which performs well here since blood pressure is generated as a linear combination of sodium intake, age, and urinary protein).

Using this model-assisted estimator with proper covariate adjustment yields:

$$\mathbb{E}[Y(1) - Y(0)] = 0.856$$

resulting in a relative error of[cite: 10]:

$$\%\text{error} = \frac{|0.856 - 1.05|}{1.05} \times 100\% = 18\%$$

2.5.2 The Peril of Naive Estimation

By contrast, if we were to ignore the confounders $\mathbf{Z}$ and naively regress Y solely on X (daily sodium intake), we would obtain:

$$\mathbb{E}[Y(1) - Y(0)] = 5.37$$

which translates to a massive relative error of:

$$\%\text{error} = \frac{|5.37 - 1.05|}{1.05} \times 100\% = 411\%$$

This striking discrepancy highlights why proper conditioning via the adjustment formula is critical to prevent severe confounding bias in observational data.